The chain rule and the Jacobian: how a slope travels backwards through a model made of many steps

A model is a long chain of steps from the input to the loss. How does the slope of the loss reach a number buried many steps deep?

Mathematics for AI and Machine Learning | Session 9 of 18 | Calculus and optimisation
KLAi Professional Learning | Jotham Martin Wambi

Today in one view: what you will leave with

By the end of today you can



Differentiate a function of a function with the chain rule and check the answer with a finite difference



Draw a computation graph, then compute forward values and backward derivatives on a tiny two-step graph



Write down and compute the Jacobian matrix of a vector function with two inputs and two outputs

Your two hours: chain rule, graphs, the Jacobian, then the lab

0:00	10	Recall, and a real question from AI that today answers
0:10	35	Concept: a picture, then a hand example
0:45	10	Break
0:55	45	Lab in Python or a spreadsheet: predict, run, change one thing
1:40	15	Where it appears in ChatGPT, Gemini, Claude; exit check
1:55	5	Save the work

Pictures and hand examples first, then the lab on the same numbers in Python or a spreadsheet.

Three questions from last time

1. What is a partial derivative?

2. What is the gradient, and which way does it point?

3. The loss of the line y-hat = w x + b at w = 1, b = 0 has gradient (-18.5, -6.5). What does it tell you to do?



Answer aloud first. Then we check together.

Answers to the three questions from Session 8

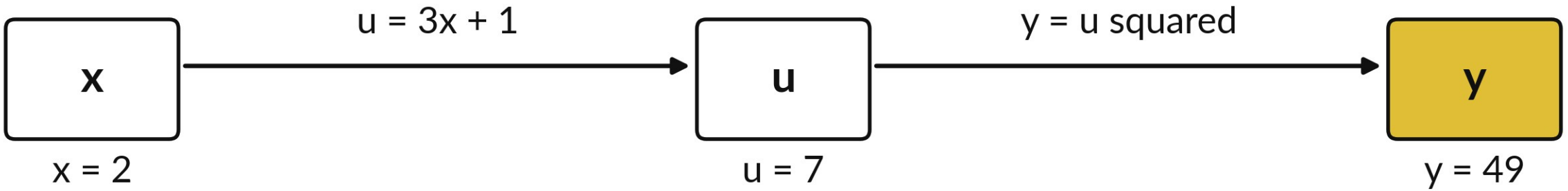
Question	Short answer
What is a partial derivative?	The slope of a function when you change one input and hold all the other inputs fixed.
What is the gradient, and which way does it point?	The vector of all the partial derivatives. It points in the direction of steepest increase, so training steps along minus the gradient.
The loss of the line y-hat = w x + b at w = 1, b = 0 has gradient (-18.5, -6.5). What does it tell you to do?	Raise both w and b, because both entries are negative. The w entry is larger in size (18.5 against 6.5), so the loss is more sensitive to w here.

If one answer surprised you, look back at Session 8 before the lab.

Words we use today

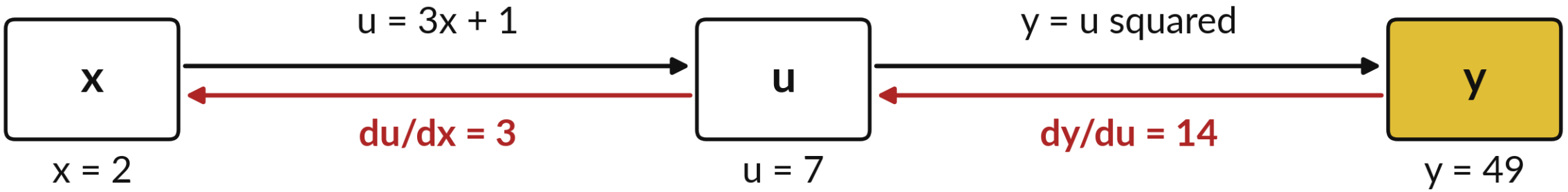
	Word	What it means
	**Composite function**	A function of a function: the output of one function becomes the input of the next. Written f(g(x)).
	**Chain rule**	The derivative of a composite function is the derivative of the outer function times the derivative of the inner function.
	**Computation graph**	A picture of a calculation as boxes (values) and arrows (operations), from the inputs to the loss.
	**Jacobian**	The matrix of all the partial derivatives of a function with several inputs and several outputs. One row for each output, one column for each input.
	**Backpropagation**	Computing the derivatives of the loss by applying the chain rule backwards through the computation graph.

A function of a function: the output of one step becomes the input of the next



Two steps. The input x = 2 becomes u = 7, and u becomes y = 49.

The chain rule: slopes along the chain multiply



A tiny step in x moves u 3 times as far; in u, y 14 times as far. So y moves 3 × 14 = 42 times as far.

The chain rule: dy/dx is dy/du times du/dx

Chain rule

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Our data at x = 2

$$\frac{dy}{dx} = 14 \times 3 = 42$$

Symbol	What it means	In our data
y = f(u)	The outer function: it takes u and returns y.	y = u squared; dy/du = 2u = 14 at u = 7
u = g(x)	The inner function: it takes x and returns u.	u = 3x + 1; du/dx = 3
dy/dx	The slope of the whole chain: how fast y changes when x changes.	42 at x = 2

Evaluate dy/du at the value of u that the forward step produced (u = 7), not at x.

Check the chain rule with a finite difference: y = (3x + 1) squared at x = 2

Step h	y(2 + h)	y(2)	Forward slope = (y(2 + h) minus y(2)) / h
1	100	49	51.0000
0.1	53.29	49	42.9000
0.01	49.4209	49	42.0900
0.001	49.042	49	42.0090

The slopes settle on 42, the chain rule answer. The central difference with h = 0.001 gives 42.0000. In the lab you check every chain rule answer this way.

Three more chain rule examples, worked with numbers at x = 1

Function y(x)	Inner u	Outer in u	du/dx	dy/du	dy/dx at x = 1
y = (3x + 1) squared	3x + 1	u squared	3	2u = 8	24
y = e to the 2x	2x	e to the u	2	e^u = 7.389	14.778
y = ln(x squared + 1)	x squared + 1	ln u	2x = 2	1/u = 0.5	1

The second row explains the remark in Session 7: e to the 2x has slope 2 times e to the 2x.

Two mistakes to avoid with the chain rule

Forgetting the inner slope

The slope of u squared is 2u.

But u = 3x + 1 also changes with x.

Multiply by du/dx = 3 as well.

Adding instead of multiplying

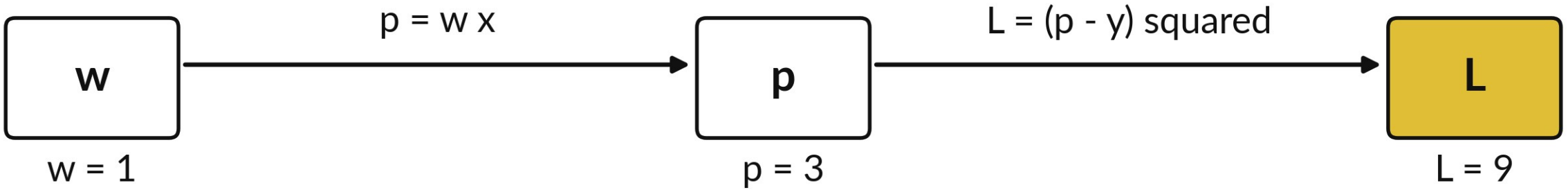
Slopes along a chain multiply.

14 + 3 = 17 is wrong here.

14 times 3 = 42 is right.

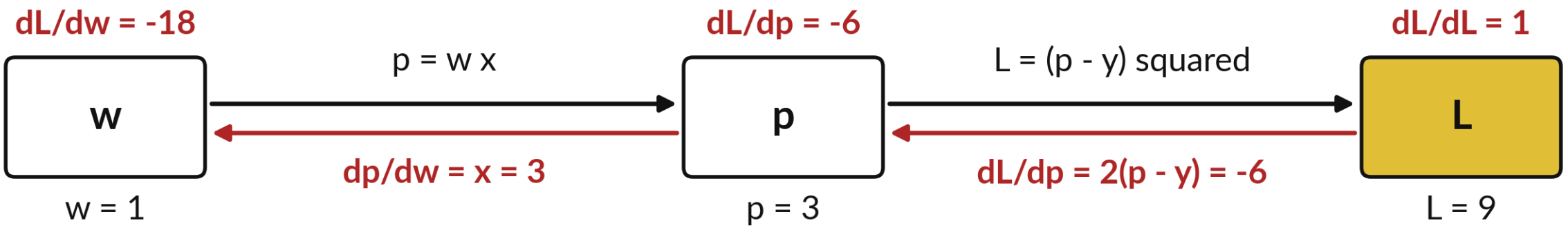
Outer slope times inner slope, each evaluated at its own value.

A computation graph draws a calculation as boxes for values and arrows for operations



One delivery: x = 3 km, y = 6. Forward: p = 1 times 3 = 3, then L = (3 - 6) squared = 9.

Backward: each box gets the slope of the loss with respect to it



Black: forward values. Red: backward derivatives. Start at 1, multiply by each local slope.

Backward pass on the tiny graph: multiply the local slopes along the path

Chain rule on the graph

$$\frac{dL}{dw} = \frac{dL}{dp} \cdot \frac{dp}{dw}$$

Our data

$$\frac{dL}{dw} = (-6) \times 3 = -18$$

Symbol	What it means	In our data
dL/dp	Local slope of the loss in p: 2 (p - y), with p = 3, y = 6.	-6
dp/dw	Local slope of p in w: p = w x, so the slope is x.	3
dL/dw	The slope of the loss in w: the product of the two.	-18

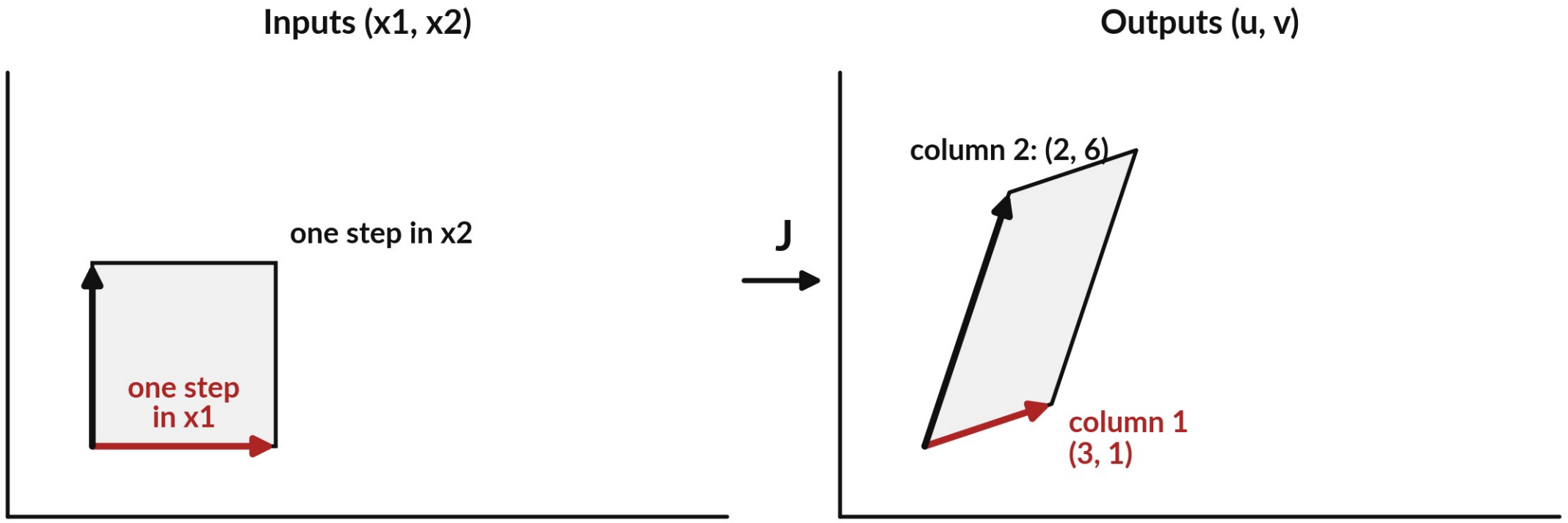
A local slope is the derivative of one box's output with respect to its own input.

Backward values for all four deliveries at w = 1: the average is the Session 8 slope

Row i	Distance x	Error p minus y	dL/dp = 2 e	dL/dw = x times dL/dp
1	1	-2	-4	-4
2	2	-3	-6	-12
3	3	-3	-6	-18
4	4	-5	-10	-40
Mean				-18.5

Row 3 is the graph we just drew: -6 and -18. The mean loss has the mean of the four backward values: -18.5, as in Session 8.

A vector function has several outputs: the Jacobian holds every slope



A very small square becomes almost a parallelogram with the columns of J as sides.

The Jacobian: one row for each output, one column for each input

Jacobian entry

$$J_{ij} = \frac{\partial F_i}{\partial x_j}$$

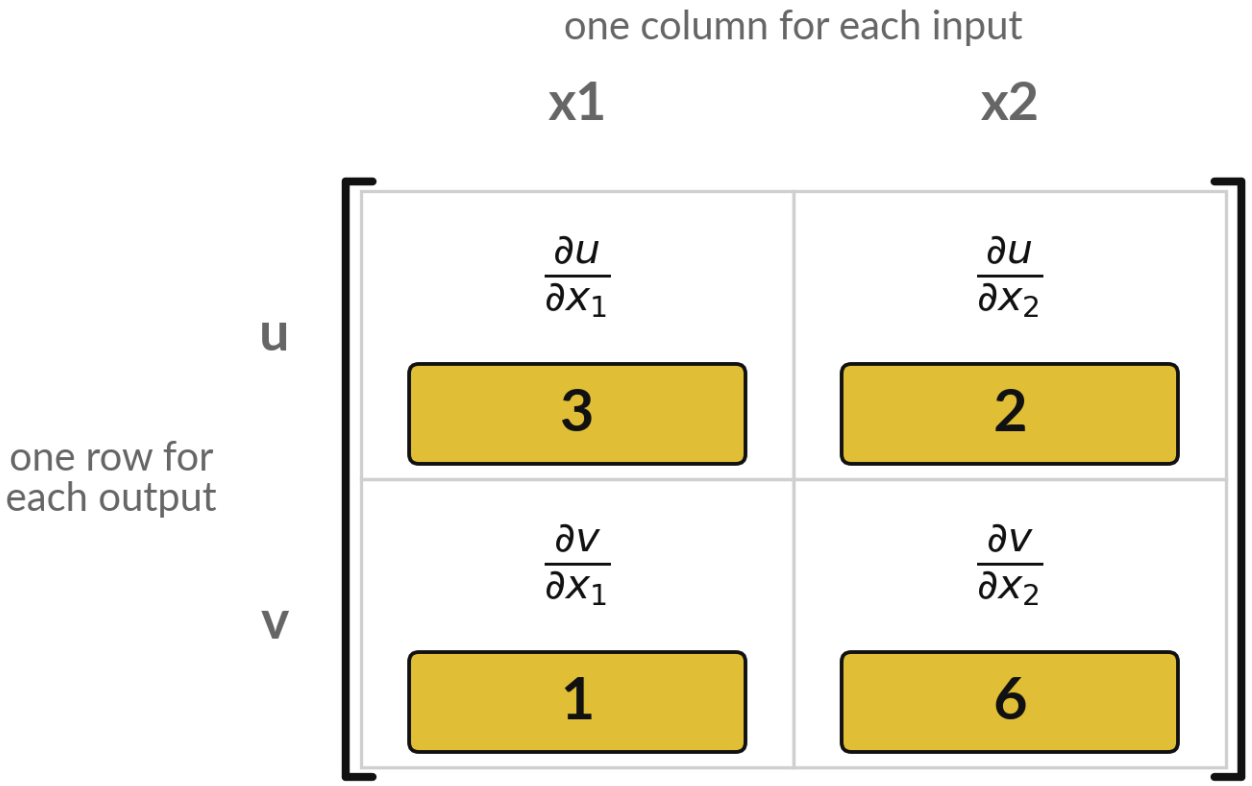
Our function

$$u = x_1 x_2 \qquad v = x_1 + x_2^2$$

Symbol	What it means	In our data
F_i	Output i of the function F. Here F_1 = u, F_2 = v.	at (2, 3): u = 6, v = 11
x_j	Input number j. Here x_1 and x_2.	the point (2, 3)
J_ij	The slope of output i in input j. J has 2 rows and 2 columns.	J = rows (3, 2) and (1, 6)

Row 1: du/dx1 = x2 = 3, du/dx2 = x1 = 2. Row 2: dv/dx1 = 1, dv/dx2 = 2 x2 = 6.

The Jacobian drawn as a matrix, with the numbers at the point (2, 3)



Each gold number is one partial derivative at the point (2, 3).

Row u: x2 = 3 and x1 = 2.

Row v: 1 and 2 x2 = 6.

Check one entry: nudging x2 from 3 to 3.001 raises v by 0.006001, so 6.001 per unit.

Check the Jacobian with finite differences: nudge one input at a time

Nudge	u changes to	v changes to	Column of J
x1: 2 + h and 2 - h, h = 0.00001	6.00003 and 5.99997	11.00001 and 10.99999	(3.000, 1.000)
x2: 3 + h and 3 - h, h = 0.00001	6.00002 and 5.99998	11.00006 and 10.99994	(2.000, 6.000)

Each column is (change in u, change in v) divided by 2h. Two inputs, two nudges, two columns.

The chain rule with a Jacobian: the backward step multiplies by J transposed

Backward through a vector step

$$\nabla_x s = J^T \nabla_{(u,v)} s$$

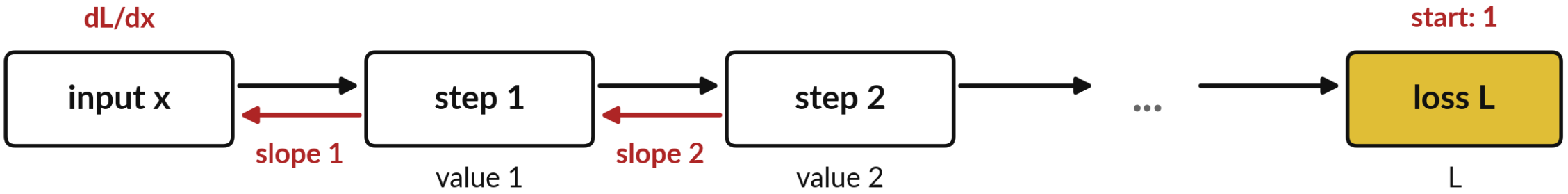
Our data for s = u v

$$\nabla_x s = (3 \cdot 11 + 1 \cdot 6,\ 2 \cdot 11 + 6 \cdot 6) = (39,\ 58)$$

Symbol	What it means	In our data
s	A single number computed from the outputs: s = u v = 6 times 11 = 66.	ds/du = v = 11; ds/dv = u = 6
J transposed	The Jacobian with rows and columns swapped. It sends output slopes back to the inputs.	J^T = rows (3, 1) and (2, 6)
gradient of s in x	The slope of s in each input, by the chain rule.	(39, 58)

Check directly: s = x1 squared x2 + x1 x2 cubed, so ds/dx1 = 39 and ds/dx2 = 58.

Many steps: the backward value is the product of the local slopes along the whole chain



Start at the loss with 1. Going back, each step multiplies by its local slope (red).

Read the recipe before you write it: forward, backward and a check

```
p = w * x
L = (p - y) ** 2
dL_dp = 2 * (p - y)
dL_dw = dL_dp * x
print(p, L, dL_dp, dL_dw)
```

```
3.0 9.0 -6.0 -18.0
```

In plain words

Lines 1 and 2: the forward pass. p is the prediction, L the loss.

Line 3: the local slope of the loss in p, 2 (p - y).

Line 4: times the local slope of p in w, which is x. This is the chain rule.

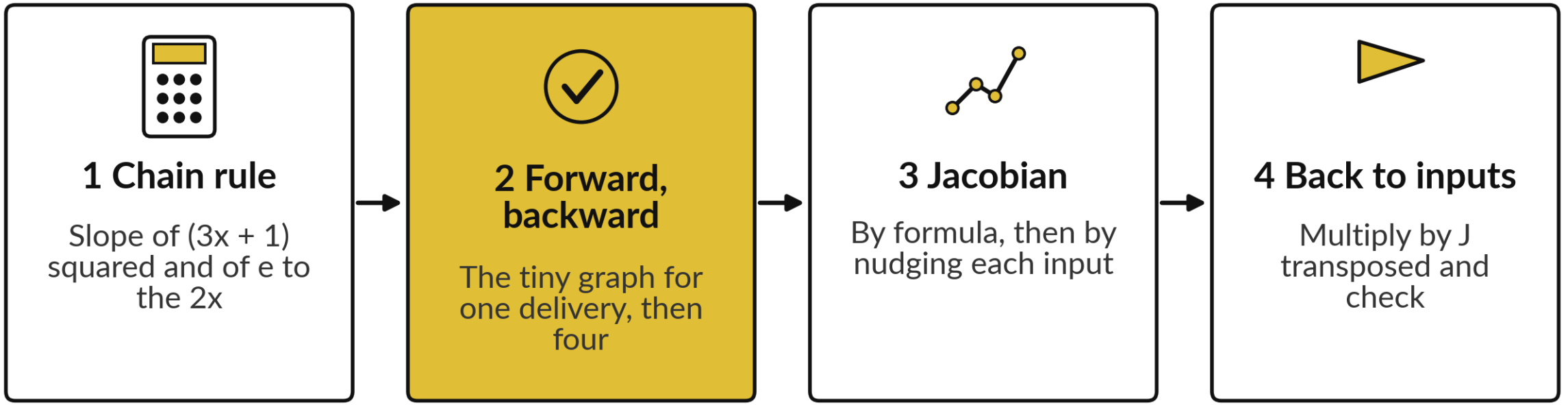
With w = 1, x = 3, y = 6 it prints 3, 9, -6 and -18.



Part 2: the lab

Close the slides. Open the notebook or the workbook.

The lab has four stages, in Python and in the spreadsheet



Python: Steps 1 to 6 of the notebook. Spreadsheet: sheets Chain, Backward and Jacobian. Predict first, then run.

Check your numbers against these before you move on

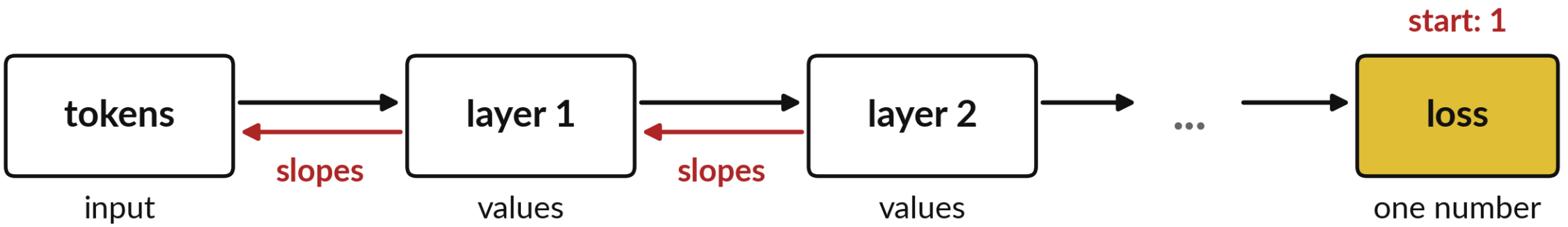
	What you compute	You should see
	Chain rule slope of (3x + 1) squared at x = 2	42
	Chain rule slope of e to the 2x at x = 1	14.778
	Forward values for the delivery x = 3, y = 6 at w = 1: p and L	3 and 9
	Backward values: dL/dp and dL/dw	-6 and -18
	Mean of the four dL/dw values at w = 1	-18.5
	Jacobian of (x1 x2, x1 + x2 squared) at (2, 3)	rows (3, 2) and (1, 6)
	Gradient of s = u v in the inputs, J transposed times (11, 6)	(39, 58)

If your number differs, stop and ask. Do not change a number to match. Find the step that went wrong.

Errors you will meet, what they mean and what to do

	What you see	What it means	What to do
	**NameError in Python**	A cell that defines a name was skipped	Run all cells from the top, in order
	**Chain slope is 14 instead of 42**	The inner slope du/dx was left out	Multiply by 3 as well as by 2u
	**Jacobian looks flipped**	Rows and columns swapped: outputs are rows	Row i is output i, column j is input j
	**Spreadsheet shows #VALUE!**	A range with text or a gap in a formula	Check that every cell in the range holds a number

Where this appears in AI: the chain rule is used once for every layer



Red: backward slopes. Meta's Llama 3 paper (2024) lists 32 and 80 transformer layers (8B, 70B).

Where this appears in AI: libraries run the chain rule for you, and you can check them



One entry for every parameter of the model

N = number of parameters

Backward pass: N slopes

PyTorch and JAX record the computation graph and run the chain rule backwards for you.

This is automatic differentiation. Nudge checks like today's test it.

A layer's Jacobian can have millions of entries, so libraries never build it. They multiply slopes by J transposed directly.

Check an AI assistant's answer with your own numbers

8 minutes

Hands-on, with any assistant

Ask an assistant for the Jacobian of F(x1, x2) = (x1 x2, x1 + x2 squared) at the point (2, 3).

Ask it to explain each entry.

Compare with ours: rows (3, 2) and (1, 6).

Run your lab Jacobian code.

If it differs, find the first entry that differs.

You keep

One line: its Jacobian, yours, match or not. Never paste confidential data into an assistant.

Exit check: three questions with short answers

Question	Short answer
What is the slope of y = (2x + 5) cubed at x = 0?	150. Outer: 3u squared = 75. Inner: 2. Product 150.
A chain has three steps with local slopes 2, -3 and 0.5. What is the slope of the output with respect to the input?	2 times -3 times 0.5 = -3.
What is the shape of the Jacobian of F(x1, x2) = (2 x1, x1 + x2), and what are its entries?	2 rows (outputs) by 2 columns (inputs): rows (2, 0) and (1, 1).

Write your own answer before you read ours. The automatic check counts the lab numbers.

Save the lab and keep the chain rule slope, the backward value and the Jacobian

What you keep from today

Keep your saved lab. Write down the chain rule slope 42, the backward value -18 for the tiny graph, and the Jacobian, rows (3, 2) and (1, 6).

Our question today

A model is a long chain of steps from the input to the loss. How does the slope of the loss reach a number buried many steps deep?

Next session

Session 10: gradient descent.

Take away

Saved lab
Notebook or workbook

Chain rule slope
42

Backward value
dL/dw = -18

Next session
Gradient descent